A COUNTEREXAMPLE TO UNIFORM L^2 STABILITY IN THE BRASCAMP–LIEB INEQUALITY

Abstract. We write down a short paper whose mathematical content matches the Lean development in the directory `Lean/`. The argument produces an explicit one-dimensional family (ϕ_S, f_S) for which

$$\frac{\mathrm{dist}_{L^2(\rho_{\phi_S})}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} = S\big(1 + O(S^{-1})\big), \qquad S \to \infty,$$

where $\delta_{\mathrm{BL},\phi}$ denotes the Brascamp–Lieb deficit and $\mathcal{O}_{\mathrm{BL},\phi} = \mathrm{span}\{1, \phi'\}$ is the optimizer space in dimension one. In particular no finite L^2-stability constant can hold uniformly along this family. We then lift the example to every dimension $d \geq 1$ by the product potential

$$\Phi_{S,d}(x, y) = \phi_S(x) + \frac{|y|^2}{2},$$

and obtain the exact higher-dimensional statement encoded in the Lean theorem: for each fixed $d \geq 1$ there is no finite constant C such that

$$\mathrm{dist}_{L^2(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^2 \leq C^2\, \delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})$$

for every parameter S for which the deficit is positive.

1. Statement

Let $\phi : \mathbb{R}^d \to \mathbb{R}$ be smooth and strictly convex, with $0 < \int_{\mathbb{R}^d} e^{-\phi(x)}\, dx < \infty$. We write

$$Z_\phi := \int_{\mathbb{R}^d} e^{-\phi(x)}\, dx, \qquad d\rho_\phi := Z_\phi^{-1} e^{-\phi(x)}\, dx.$$

For a locally Lipschitz $f \in L^2(\rho_\phi)$, the Brascamp–Lieb energy and deficit are

$$\mathcal{E}_\phi(f) := \int_{\mathbb{R}^d} \langle (D^2\phi)^{-1} \nabla f, \nabla f \rangle\, d\rho_\phi, \qquad \delta_{\mathrm{BL},\phi}(f) := \mathcal{E}_\phi(f) - \mathrm{Var}_{\rho_\phi}(f).$$

The optimizer space is

$$\mathcal{O}_{\mathrm{BL},\phi} := \{a \cdot \nabla\phi + b : \ a \in \mathbb{R}^d, \ b \in \mathbb{R}\}.$$

The concrete construction below yields a one-dimensional family (ϕ_S, f_S) with $\delta_{\mathrm{BL},\phi_S}(f_S) > 0$ for all sufficiently large S and

$$\frac{\mathrm{dist}_{L^2(\rho_{\phi_S})}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} = S\big(1 + O(S^{-1})\big).$$

This already implies that any stability statement of the form

$$\mathrm{dist}_{L^2(\rho_\phi)}(f, \mathcal{O}_{\mathrm{BL},\phi}) \leq C\, \delta_{\mathrm{BL},\phi}(f)^{1/2}$$

must fail on every class of admissible one-dimensional data that contains the family (ϕ_S, f_S).

The Lean build also proves the following exact fixed-dimension product-family statement.

Theorem 1.1. *For every integer $d \geq 1$ there exists a smooth strictly convex family $\Phi_{S,d} : \mathbb{R}^d \to \mathbb{R}$ and a locally Lipschitz family $F_{S,d} \in L^2(\rho_{\Phi_{S,d}})$ such that:*

1. *for all sufficiently large S one has $\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}) > 0$;*
2. *for every finite $C \in \mathbb{R}$ there exists S with $\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}) > 0$ and*

$$\mathrm{dist}_{L^2(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^2 > C^2\, \delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}).$$

Equivalently, for fixed $d \geq 1$ there is no finite constant C such that

$$\mathrm{dist}_{L^2(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^2 \leq C^2\, \delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})$$

for every parameter S with $\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}) > 0$.

The proof is one-dimensional at heart. The higher-dimensional statement is obtained by tensoring with a standard Gaussian in the transverse variables.

2. The one-dimensional potential

Fix a non-negative even C^∞ bump $\kappa : \mathbb{R} \to [0, \infty)$ supported in $[-1, 1]$ and normalized by

$$\int_{\mathbb{R}} \kappa(u)\, du = 1.$$

For parameters $S \geq 1$, $\varepsilon \in (0, \frac{1}{4})$ and $\eta > 0$, define

$$\phi''_{S,\varepsilon,\eta}(x) := \eta + \frac{S}{\varepsilon}\kappa\left(\frac{x-1}{\varepsilon}\right) + \frac{S}{\varepsilon}\kappa\left(\frac{x+1}{\varepsilon}\right), \tag{1}$$

and impose the normalization

$$\phi_{S,\varepsilon,\eta}(0) = 0, \qquad \phi'_{S,\varepsilon,\eta}(0) = 0.$$

The right-hand side of (1) is smooth, even, and bounded below by η, so $\phi_{S,\varepsilon,\eta}$ is smooth, even, and strictly convex.

We make the specific choice

$$\varepsilon = S^{-3}, \qquad \eta = S^{-4}, \tag{2}$$

and abbreviate

$$\phi_S := \phi_{S,S^{-3},S^{-4}}, \qquad \rho_S := \rho_{\phi_S}, \qquad Z_S := \int_{\mathbb{R}} e^{-\phi_S(x)}\, dx.$$

All asymptotic notation below refers to the limit $S \to \infty$.

We decompose $\mathbb{R}$ into the central core, two transition layers, and the exterior:

$$C_S := [-1+\varepsilon, 1-\varepsilon], \qquad I_S^+ := (1-\varepsilon, 1+\varepsilon), \qquad I_S^- := (-1-\varepsilon, -1+\varepsilon),$$

$$T_S := I_S^+ \cup I_S^-, \qquad E_S := \mathbb{R} \setminus (C_S \cup T_S).$$

The next lemma summarizes the explicit shape of ϕ_S needed in the sequel.

Lemma 2.1. *For the choice (2), the following hold for all sufficiently large S:*

(1) *on C_S one has*

$$\phi''_S \equiv \eta, \qquad \phi'_S(x) = \eta x, \qquad \phi_S(x) = \frac{\eta x^2}{2} \in \left[0, \frac{\eta}{2}\right];$$

(2) *on T_S one has $\phi_S(x) = O(S^{-2})$ uniformly, hence $e^{\phi_S(x)} = 1 + O(S^{-2})$ uniformly on T_S;*

(3) *for $x > 1 + \varepsilon$,*

$$\phi'_S(x) = S + \eta x,$$

and therefore

$$\phi_S(x) = \phi_S(1+\varepsilon) + S(x-1-\varepsilon) + \frac{\eta}{2}\big(x^2 - (1+\varepsilon)^2\big);$$

(4) *by evenness, the corresponding formulas on the left tail follow from the right-tail ones by the substitution $x \mapsto -x$.*

Proof. On the core C_S both translated bumps in (1) vanish, so $\phi_S'' = \eta$ there. Integrating and using $\phi_S'(0) = \phi_S(0) = 0$ gives the displayed core formulas.

On I_S^+ we have

$$\phi_S(x) = \phi_S(1-\varepsilon) + \int_{1-\varepsilon}^{x} \phi_S'(t)\,dt.$$

Since $\phi_S(1-\varepsilon) \le \eta/2$ and $\sup_{I_S^+} \phi_S' \le \eta + S$, we obtain

$$\phi_S(x) \le \frac{\eta}{2} + 2\varepsilon(\eta + S) = O(S^{-2}),$$

using (2). The same estimate holds on I_S^- by evenness. The bound on $e^{\phi_S} - 1$ is immediate.

On $(1+\varepsilon, \infty)$ both bumps have again vanished, so $\phi_S'' = \eta$. Since the total mass of the right bump is S, $\phi_S'(1+\varepsilon) = S + \eta(1+\varepsilon)$, and the right-tail formulas follow by integration. The left tail is obtained by symmetry. □

3. Normalization and the tail mass

The next lemma is the only normalization input needed later on.

Lemma 3.1. *With the choice* (2)*, as* $S \to \infty$*:*

(1)
$$\int_{1+\varepsilon}^{\infty} e^{-\phi_S(x)}\,dx = \frac{1}{S} + O(S^{-3});$$

(2)
$$Z_S = 2 + \frac{2}{S} + O(S^{-3});$$

(3) *if*
$$q_S := \rho_S(E_S) = \rho_S(\{|x| > 1+\varepsilon\}), \qquad t_S := \rho_S(T_S),$$
then
$$q_S = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}), \qquad t_S = O(S^{-3}).$$

Proof. By Lemma 2.1, for $x > 1+\varepsilon$ we have

$$\phi_S(x) = \phi_S(1+\varepsilon) + \widetilde{S}\,(x-1-\varepsilon) + \frac{\eta}{2}(x-1-\varepsilon)^2,$$

where $\widetilde{S} = S + \eta(1+\varepsilon) = S + O(S^{-4})$. Thus

$$\int_{1+\varepsilon}^{\infty} e^{-\phi_S(x)}\,dx = e^{-\phi_S(1+\varepsilon)} \int_0^{\infty} e^{-\widetilde{S}u - \eta u^2/2}\,du.$$

Since $\phi_S(1+\varepsilon) = O(S^{-2})$, the prefactor equals $1 + O(S^{-2})$. Also

$$0 \le \frac{1}{\widetilde{S}} - \int_0^{\infty} e^{-\widetilde{S}u - \eta u^2/2}\,du \le \frac{\eta}{2}\int_0^{\infty} u^2 e^{-\widetilde{S}u}\,du = \frac{\eta}{\widetilde{S}^3} = O(S^{-7}),$$

so the tail integral is $1/S + O(S^{-6})$. This proves (1).

For (2), split $\int_0^{\infty} e^{-\phi_S}\,dx$ into the core, the right layer, and the right tail. On the core, $\phi_S(x) = \eta x^2/2$, hence

$$\int_0^{1-\varepsilon} e^{-\phi_S(x)}\,dx = 1 - \varepsilon + O(\eta).$$

On the layer I_S^+, Lemma 2.1 gives $e^{-\phi_S(x)} = 1 + O(S^{-2})$ uniformly, so

$$\int_{1-\varepsilon}^{1+\varepsilon} e^{-\phi_S(x)}\,dx = 2\varepsilon + O(S^{-5}).$$

On the tail we use (1). Therefore

$$\int_0^\infty e^{-\phi_S(x)}\,dx = 1 + \frac{1}{S} + O(S^{-3}),$$

and by evenness

$$Z_S = 2 + \frac{2}{S} + O(S^{-3}).$$

Finally,

$$t_S \le \frac{|T_S|}{Z_S} \sup_{x\in T_S} e^{-\phi_S(x)} = O(\varepsilon) = O(S^{-3}).$$

Moreover,

$$q_S = \frac{2}{Z_S}\int_{1+\varepsilon}^\infty e^{-\phi_S(x)}\,dx = \frac{2/S + O(S^{-3})}{2 + 2/S + O(S^{-3})} = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}),$$

which proves (3). □

4. The transition test function

Define

$$g_S(x) := \begin{cases} 0, & x \in C_S, \\ \dfrac{\int_{1-\varepsilon}^{|x|} \phi_S''(t)e^{\phi_S(t)}\,dt}{\int_{I_S^+} \phi_S''(t)e^{\phi_S(t)}\,dt}, & |x| \in (1-\varepsilon, 1+\varepsilon), \\ 1, & x \in E_S. \end{cases} \tag{3}$$

This function is even, continuous, bounded between 0 and 1, and globally Lipschitz.

Set

$$A_S := \int_{I_S^+} \phi_S''(t)e^{\phi_S(t)}\,dt.$$

The energy of g_S is explicit.

Lemma 4.1. *As $S \to \infty$,*

$$A_S = S\big(1 + O(S^{-2})\big),$$

and

$$\mathcal{E}_{\phi_S}(g_S) = \frac{2}{Z_S A_S} = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}).$$

Proof. By Lemma 2.1, $e^{\phi_S} = 1 + O(S^{-2})$ uniformly on I_S^+, while

$$\int_{I_S^+} \phi_S''(t)\,dt = S + 2\eta\varepsilon = S + O(S^{-7}).$$

Hence

$$A_S = \int_{I_S^+} \phi_S''(t)e^{\phi_S(t)}\,dt = S\big(1 + O(S^{-2})\big).$$

To compute the energy, note that g_S is constant on $C_S \cup E_S$, so only the two layers contribute. On I_S^+ one has

$$g_S'(x) = \frac{\phi_S''(x)e^{\phi_S(x)}}{A_S},$$

and the same formula holds on I_S^- up to the sign change forced by evenness. Therefore

$$\mathcal{E}_{\phi_S}(g_S) = \frac{2}{Z_S}\int_{I_S^+} \frac{(g_S'(x))^2}{\phi_S''(x)} e^{-\phi_S(x)}\,dx = \frac{2}{Z_S A_S}.$$

Combining this with Lemma 3.1 gives

$$\frac{2}{Z_S A_S} = \frac{2}{(2+2/S+O(S^{-3}))\,S(1+O(S^{-2}))} = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}).$$

□

Lemma 4.2. *The variance of g_S satisfies*

$$\mathrm{Var}_{\rho_S}(g_S) = q_S(1-q_S) + O(S^{-3}) = \frac{1}{S} - \frac{2}{S^2} + O(S^{-3}).$$

Proof. Since $g_S = 0$ on C_S, $g_S = 1$ on E_S, and $0 \le g_S \le 1$ on T_S, we have

$$\int g_S \, d\rho_S = q_S + O(t_S), \qquad \int g_S^2 \, d\rho_S = q_S + O(t_S).$$

By Lemma 3.1, $t_S = O(S^{-3})$, so

$$\mathrm{Var}_{\rho_S}(g_S) = \int g_S^2 \, d\rho_S - \left(\int g_S \, d\rho_S\right)^2 = q_S - q_S^2 + O(S^{-3}).$$

Substituting the expansion for q_S from Lemma 3.1 gives the result. □

Now define

$$c_S := \int_{\mathbb{R}} g_S \, d\rho_S, \qquad f_S := g_S - c_S.$$

Then f_S is even and has ρ_S-mean zero. Since the deficit and the variance are invariant under addition of constants,

$$\mathcal{E}_{\phi_S}(f_S) = \mathcal{E}_{\phi_S}(g_S), \qquad \mathrm{Var}_{\rho_S}(f_S) = \mathrm{Var}_{\rho_S}(g_S), \qquad \delta_{\mathrm{BL},\phi_S}(f_S) = \delta_{\mathrm{BL},\phi_S}(g_S). \tag{4}$$

Lemma 4.3. *The function f_S is orthogonal in $L^2(\rho_S)$ to*

$$\mathcal{O}_{\mathrm{BL},\phi_S} = \mathrm{span}\{1, \phi_S'\}.$$

Consequently

$$\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2 = \|f_S\|^2_{L^2(\rho_S)} = \mathrm{Var}_{\rho_S}(g_S).$$

Proof. By construction $\int f_S \, d\rho_S = 0$, so $f_S \perp 1$. Also f_S is even, ϕ_S' is odd, and $e^{-\phi_S}$ is even, hence $f_S(x)\phi_S'(x)e^{-\phi_S(x)}$ is odd and integrable. Therefore

$$\int_{\mathbb{R}} f_S(x)\phi_S'(x) \, d\rho_S(x) = 0.$$

Thus $f_S \perp \mathrm{span}\{1, \phi_S'\}$. The distance identity follows because an orthogonal projection onto $\mathcal{O}_{\mathrm{BL},\phi_S}$ vanishes on f_S, while

$$\|f_S\|^2_{L^2(\rho_S)} = \mathrm{Var}_{\rho_S}(g_S)$$

since f_S is the centered version of g_S. □

Proposition 4.4. *As $S \to \infty$,*

$$\mathcal{E}_{\phi_S}(f_S) = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}), \qquad \mathrm{Var}_{\rho_S}(f_S) = \frac{1}{S} - \frac{2}{S^2} + O(S^{-3}), \qquad \delta_{\mathrm{BL},\phi_S}(f_S) = \frac{1}{S^2} + O(S^{-3}).$$

In particular, $\delta_{\mathrm{BL},\phi_S}(f_S) > 0$ for all sufficiently large S.

Proof. The first identity follows from Lemma 4.1 and (4). The second follows from Lemma 4.2 and (4). Subtracting gives

$$\delta_{\mathrm{BL},\phi_S}(f_S) = \left(\frac{1}{S} - \frac{1}{S^2}\right) - \left(\frac{1}{S} - \frac{2}{S^2}\right) + O(S^{-3}) = \frac{1}{S^2} + O(S^{-3}),$$

and the eventual positivity is immediate. □

5. The one-dimensional contradiction

Theorem 5.1. *The family (ϕ_S, f_S) satisfies*

$$\frac{\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} = S\big(1 + O(S^{-1})\big), \qquad S \to \infty.$$

In particular, for every finite C there exists S such that $\delta_{\mathrm{BL},\phi_S}(f_S) > 0$ and

$$\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2 > C^2\, \delta_{\mathrm{BL},\phi_S}(f_S).$$

Proof. By Lemma 4.3 and Proposition 4.4,

$$\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2 = \frac{1}{S} - \frac{2}{S^2} + O(S^{-3}),$$

while

$$\delta_{\mathrm{BL},\phi_S}(f_S) = \frac{1}{S^2} + O(S^{-3}).$$

Therefore

$$\frac{\mathrm{dist}^2}{\delta_{\mathrm{BL}}} = \frac{(1/S)\big(1 - 2/S + O(S^{-2})\big)}{(1/S^2)\big(1 + O(S^{-1})\big)} = S\big(1 + O(S^{-1})\big).$$

The second assertion follows because the ratio tends to $+\infty$ and the deficit is eventually positive. □

Remark 5.2. The Lean theorem in dimension one is phrased in a slightly more abstract way: it records only the numerical invariants "distance squared" and "deficit" of an admissible datum. Theorem 5.1 is exactly the concrete counterexample needed for that abstract formulation: any class of admissible one-dimensional Brascamp–Lieb data that contains the family (ϕ_S, f_S) cannot satisfy a uniform square-root stability estimate.

6. The higher-dimensional lift

Fix $d \geq 2$. Write points of $\mathbb{R}^d$ as $(x, y) \in \mathbb{R} \times \mathbb{R}^{d-1}$ and define

$$\Phi_{S,d}(x, y) := \phi_S(x) + \frac{|y|^2}{2}, \qquad F_{S,d}(x, y) := f_S(x).$$

Let γ_{d-1} denote the standard Gaussian probability measure on $\mathbb{R}^{d-1}$. Then

$$\rho_{\Phi_{S,d}} = \rho_S \otimes \gamma_{d-1},$$

and

$$D^2\Phi_{S,d}(x, y) = \mathrm{diag}(\phi_S''(x), I_{d-1}).$$

Hence

$$\nabla F_{S,d}(x, y) = (f_S'(x), 0), \qquad \langle (D^2\Phi_{S,d})^{-1}\nabla F_{S,d}, \nabla F_{S,d}\rangle = \frac{(f_S'(x))^2}{\phi_S''(x)}.$$

By Fubini,

$$\mathcal{E}_{\Phi_{S,d}}(F_{S,d}) = \mathcal{E}_{\phi_S}(f_S), \tag{5}$$

and likewise

$$\mathrm{Var}_{\rho_{\Phi_{S,d}}}(F_{S,d}) = \mathrm{Var}_{\rho_S}(f_S), \qquad \delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}) = \delta_{\mathrm{BL},\phi_S}(f_S). \tag{6}$$

The optimizer space is

$$\mathcal{O}_{\mathrm{BL},\Phi_{S,d}} = \mathrm{span}\{1, \phi_S'(x), y_1, \ldots, y_{d-1}\}.$$

Since $\int f_S\, d\rho_S = 0$ and each Gaussian coordinate has zero mean, $F_{S,d}$ is orthogonal to every generator of this space:

$$\int F_{S,d}\, d(\rho_S \otimes \gamma_{d-1}) = 0,$$

$$\int F_{S,d}(x,y)\phi_S'(x)\,d(\rho_S\otimes\gamma_{d-1}) = \left(\int f_S(x)\phi_S'(x)\,d\rho_S(x)\right)\left(\int d\gamma_{d-1}\right) = 0,$$

$$\int F_{S,d}(x,y)y_j\,d(\rho_S\otimes\gamma_{d-1}) = \left(\int f_S(x)\,d\rho_S(x)\right)\left(\int y_j\,d\gamma_{d-1}(y)\right) = 0.$$

Therefore

$$\operatorname{dist}_{L^2(\rho_{\Phi_{S,d}})}(F_{S,d},\mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^2 = \|F_{S,d}\|^2_{L^2(\rho_{\Phi_{S,d}})} = \|f_S\|^2_{L^2(\rho_S)}. \tag{7}$$

Theorem 6.1. *For every integer $d\geq 2$,*

$$\frac{\operatorname{dist}_{L^2(\rho_{\Phi_{S,d}})}(F_{S,d},\mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^2}{\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})} = S\big(1+O(S^{-1})\big), \qquad S\to\infty.$$

In particular, for every finite C there exists S such that $\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})>0$ and

$$\operatorname{dist}_{L^2(\rho_{\Phi_{S,d}})}(F_{S,d},\mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^2 > C^2\,\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}).$$

Proof. By (5), (6), and (7), the distance squared and deficit for $(\Phi_{S,d},F_{S,d})$ are exactly the one-dimensional ones for (ϕ_S,f_S). The claim therefore follows from Theorem 5.1. □

Proof of Theorem 1.1. For $d=1$ this is exactly Theorem 5.1. For $d\geq 2$ it is Theorem 6.1. The equivalent formulation with a universal bound for all S is just the negation of the pointwise contradiction statement. □

Remark 6.2. The argument above proves slightly more than the bare non-existence of a finite constant: it shows that along the concrete family the ratio

$$\frac{\operatorname{dist}^2_{L^2}}{\delta_{\mathrm{BL}}}$$

actually grows linearly in S. The Lean theorem packages only the exact non-existence statement from Theorem 1.1, but it is obtained from this stronger asymptotic behavior.

7. Intermediate L^p distances

The preceding sections use the Hilbert space distance because the Brascamp–Lieb variance and the optimizer space are naturally expressed in L^2. The same construction, however, also rules out the direct L^p analogue of the square-root stability estimate for every fixed $1<p<2$. The only point to check is the size of the distance to the optimizer space in L^p.

For a probability measure ν and a linear space $V\subset L^p(\nu)$, we use the notation

$$\operatorname{dist}_{L^p(\nu)}(h,V) := \inf_{v\in V}\|h-v\|_{L^p(\nu)}.$$

In the present one-dimensional example,

$$\mathcal{O}_{\mathrm{BL},\phi_S} = \operatorname{span}\{1,\phi_S'\}.$$

Thus an element of the optimizer space has the form $a\phi_S'+b$.

Lemma 7.1. *Let $1\leq p<\infty$ be fixed. Along the one-dimensional family constructed above,*

$$\operatorname{dist}_{L^p(\rho_S)}(f_S,\mathcal{O}_{\mathrm{BL},\phi_S})^p \asymp q_S, \qquad S\to\infty.$$

In particular, since $q_S\asymp S^{-1}$,

$$\operatorname{dist}_{L^p(\rho_S)}(f_S,\mathcal{O}_{\mathrm{BL},\phi_S}) \asymp S^{-1/p}.$$

Proof. Recall that

$$f_S = g_S - c_S, \qquad c_S = \int_{\mathbb{R}} g_S \, d\rho_S.$$

From Lemma 3.1 and the proof of Lemma 4.2, we have

$$c_S = q_S + O(t_S) = q_S + O(S^{-3}), \qquad q_S \asymp S^{-1}, \qquad t_S = O(S^{-3}).$$

Moreover $g_S = 0$ on C_S, $g_S = 1$ on E_S, and $0 \le g_S \le 1$ on T_S. Therefore

$$f_S = -c_S \quad \text{on } C_S, \qquad f_S = 1 - c_S \quad \text{on } E_S, \qquad |f_S| \le 1 + c_S \quad \text{on } T_S.$$

Since $\rho_S(C_S) = 1 - q_S - t_S = 1 + O(S^{-1})$, the function is very small on most of the mass and of order one on a set of mass q_S.

We first remove the odd optimizer direction. For every $a, b \in \mathbb{R}$, the function $f_S - b$ is even, while ϕ_S' is odd and ρ_S is even. Pairing the points x and $-x$ and using the convexity of $u \mapsto |u|^p$ gives

$$\frac{|f_S(x) - b - a\phi_S'(x)|^p + |f_S(x) - b + a\phi_S'(x)|^p}{2} \ge |f_S(x) - b|^p.$$

After integration,

$$\|f_S - a\phi_S' - b\|^p_{L^p(\rho_S)} \ge \|f_S - b\|^p_{L^p(\rho_S)}. \tag{8}$$

Thus the lower bound for the full optimizer space reduces to a lower bound against constants.

Fix $b \in \mathbb{R}$. If $|b + c_S| \le 1/2$, then on E_S,

$$|f_S - b| = |1 - c_S - b| = |1 - (b + c_S)| \ge \frac{1}{2},$$

and hence

$$\|f_S - b\|^p_{L^p(\rho_S)} \ge 2^{-p} q_S.$$

If instead $|b + c_S| > 1/2$, then on C_S,

$$|f_S - b| = |-c_S - b| > \frac{1}{2},$$

so

$$\|f_S - b\|^p_{L^p(\rho_S)} \ge 2^{-p} \rho_S(C_S).$$

For all sufficiently large S, $\rho_S(C_S) \ge 1/2$ and $q_S \le 1$, so this second lower bound is also bounded below by a constant multiple of q_S. Together with (8), this proves

$$\operatorname{dist}_{L^p(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^p \gtrsim q_S.$$

For the reverse inequality, take the optimizer element to be 0. Using the pointwise description above,

$$\|f_S\|^p_{L^p(\rho_S)} \le c_S^p \rho_S(C_S) + (1 - c_S)^p q_S + (1 + c_S)^p t_S.$$

Here $c_S \asymp q_S$, $0 < q_S < 1$, and $t_S = O(S^{-3}) = O(q_S)$, so the right-hand side is $O(q_S)$. Hence

$$\operatorname{dist}_{L^p(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^p \le \|f_S\|^p_{L^p(\rho_S)} \lesssim q_S.$$

The two estimates prove the lemma. □

Theorem 7.2. *Let $1 < p < 2$ be fixed. Along the one-dimensional family constructed above,*

$$\frac{\operatorname{dist}_{L^p(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})}{\delta_{\mathrm{BL},\phi_S}(f_S)^{1/2}} \asymp S^{1-1/p},$$

and in particular the ratio tends to $+\infty$ as $S \to \infty$. Equivalently,

$$\frac{\operatorname{dist}_{L^p(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} \asymp S^{2-2/p} \to +\infty.$$

Consequently there is no finite constant C such that

$$\mathrm{dist}_{L^p(\rho_\phi)}(f, \mathcal{O}_{\mathrm{BL},\phi}) \le C\, \delta_{\mathrm{BL},\phi}(f)^{1/2}$$

holds simultaneously for all one-dimensional admissible data containing the family (ϕ_S, f_S).

Proof. By Lemma 7.1,

$$\mathrm{dist}_{L^p(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S}) \asymp S^{-1/p}.$$

By Proposition 4.4,

$$\delta_{\mathrm{BL},\phi_S}(f_S)^{1/2} \asymp S^{-1}.$$

Therefore

$$\frac{\mathrm{dist}_{L^p(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})}{\delta_{\mathrm{BL},\phi_S}(f_S)^{1/2}} \asymp \frac{S^{-1/p}}{S^{-1}} = S^{1-1/p}.$$

Since $p > 1$, the exponent $1 - 1/p$ is positive, and the ratio diverges. Squaring the numerator gives the equivalent formulation with exponent $2 - 2/p$. □

Theorem 7.3. *Let $d \ge 2$ and $1 < p < 2$ be fixed. Along the product family $(\Phi_{S,d}, F_{S,d})$,*

$$\frac{\mathrm{dist}_{L^p(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}})}{\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})^{1/2}} \asymp S^{1-1/p}, \qquad S \to \infty.$$

In particular, for every finite C there exists S such that $\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d}) > 0$ and

$$\mathrm{dist}_{L^p(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}}) > C\, \delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})^{1/2}.$$

Proof. As in the L^2 product argument,

$$\mathcal{O}_{\mathrm{BL},\Phi_{S,d}} = \mathrm{span}\{1, \phi_S'(x), y_1, \dots, y_{d-1}\}.$$

Let

$$H(x, y) := F_{S,d}(x, y) - b = f_S(x) - b.$$

This function is even in x and invariant under every sign change $y_j \mapsto -y_j$. The probability measure $\rho_{\Phi_{S,d}} = \rho_S \otimes \gamma_{d-1}$ has the same symmetries, while $\phi_S'(x)$ is odd in x and each y_j is odd in its own Gaussian variable. Therefore, by averaging over the finite group generated by these sign changes and using convexity of $u \mapsto |u|^p$, for every $a \in \mathbb{R}$, $\alpha \in \mathbb{R}^{d-1}$, and $b \in \mathbb{R}$,

$$\|F_{S,d} - a\phi_S' - \alpha \cdot y - b\|^p_{L^p(\rho_{\Phi_{S,d}})} \ge \|F_{S,d} - b\|^p_{L^p(\rho_{\Phi_{S,d}})}.$$

Since $F_{S,d}$ is independent of y, the right-hand side equals

$$\|f_S - b\|^p_{L^p(\rho_S)}.$$

The lower-bound argument from Lemma 7.1, applied to constants b, gives

$$\mathrm{dist}_{L^p(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^p \gtrsim q_S.$$

Conversely, taking the zero optimizer gives

$$\mathrm{dist}_{L^p(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}})^p \le \|F_{S,d}\|^p_{L^p(\rho_{\Phi_{S,d}})} = \|f_S\|^p_{L^p(\rho_S)} \lesssim q_S.$$

Thus the product-family L^p distance has the same size as the one-dimensional one:

$$\mathrm{dist}_{L^p(\rho_{\Phi_{S,d}})}(F_{S,d}, \mathcal{O}_{\mathrm{BL},\Phi_{S,d}}) \asymp S^{-1/p}.$$

The deficit is unchanged by the product lift, by (6), so

$$\delta_{\mathrm{BL},\Phi_{S,d}}(F_{S,d})^{1/2} = \delta_{\mathrm{BL},\phi_S}(f_S)^{1/2} \asymp S^{-1}.$$

Dividing gives the claimed asymptotic ratio. □

Remark 7.4. At $p = 1$ the same computation is exactly borderline. Lemma 7.1 gives

$$\mathrm{dist}_{L^1(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S}) \asymp S^{-1},$$

while Proposition 4.4 gives

$$\delta_{\mathrm{BL},\phi_S}(f_S)^{1/2} \asymp S^{-1}.$$

Thus this counterexample does not contradict an L^1 square-root stability estimate. In fact, the companion paper [1] proves precisely such a sharp uniform L^1 stability theorem for the Brascamp–Lieb deficit. The present construction shows that the topology in that theorem cannot be strengthened to any fixed L^p topology with $p > 1$ by a dimension-free constant of the same square-root form.

References

[1] J. M. Machado and J. P. G. Ramos, *Quantitative stability for Brascamp–Lieb and moment measures*, arXiv:2511.22636v2.

[2] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log-concave functions, and with an application to the diffusion equation*, J. Funct. Anal. 22 (1976), 366–389.